IEEE/CIC ICCC

TUT-07: Generative AI for Radio Map Construction and Its Applications in Low-Altitude Economy: Principles, Datasets, and Digital Twin

Physics Informed Learning Methods for Radio Map Construction

Xiucheng Wang
Xidian University

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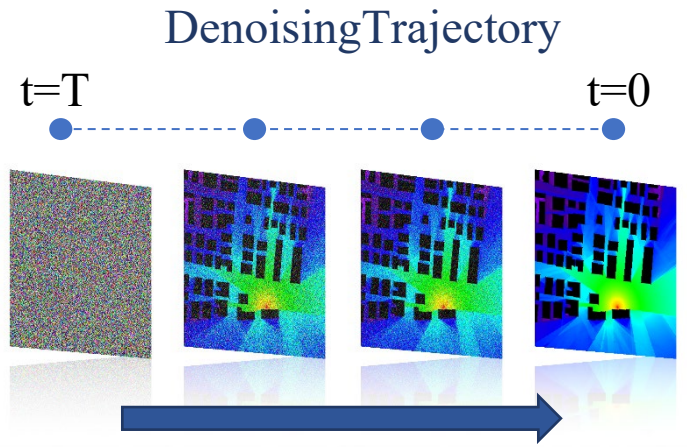
Physics-Informed Radio Map Construction

Geometry Priors, Helmholtz-Aligned Features, and Propagation-Aware Diffusion





Diffusion-Based Refinement



$$\nabla^2 u + k^2 u = 0$$

$$u(r) = Ae^{jkr}/r$$

$$x_T \quad x_t \quad x_0$$

$$dx = -\nabla E(x,t)dt + \sqrt{2\beta(t)}\,dw_t$$

Background: Environment-Aware 6G & Digital Twin

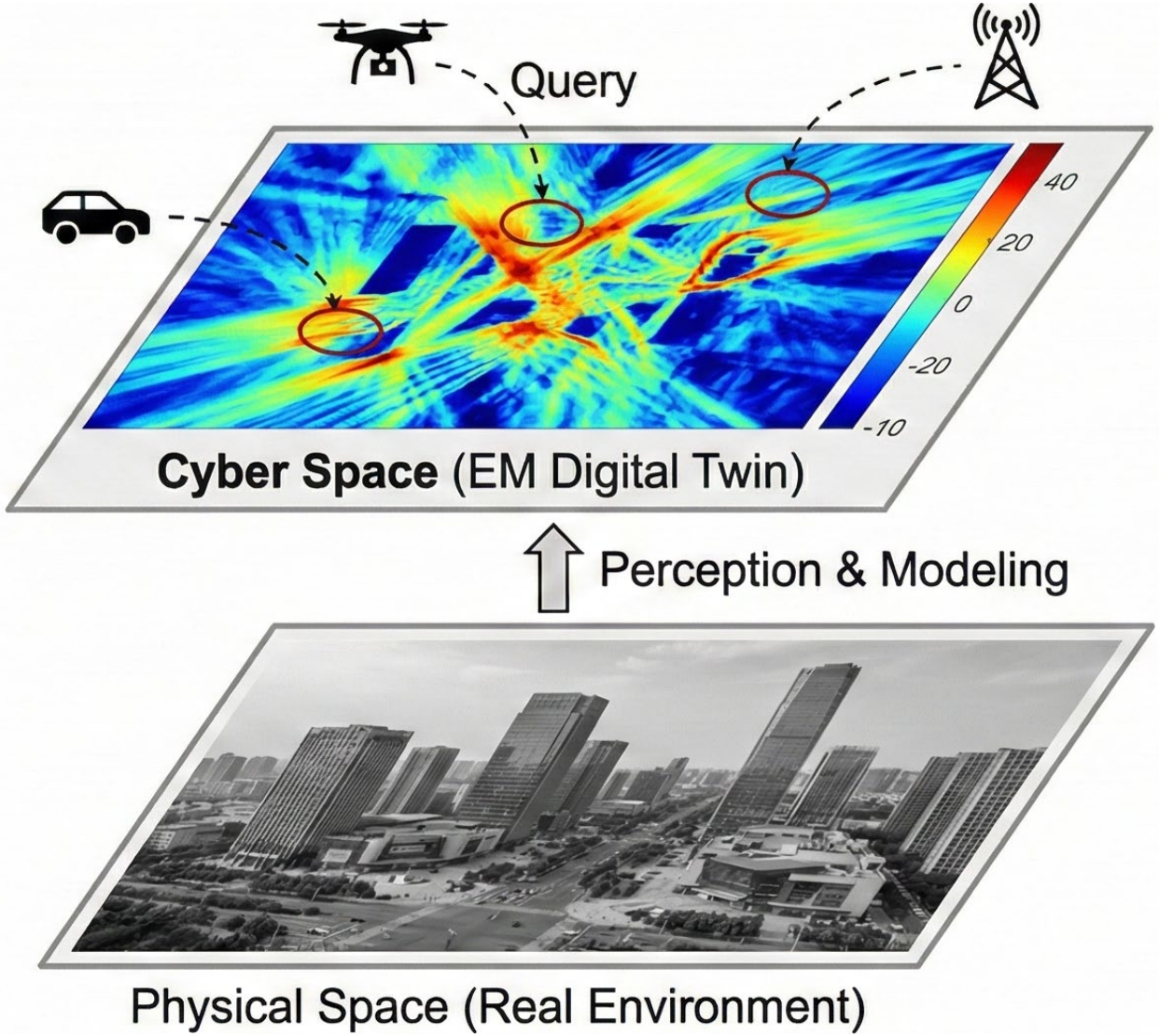
- **Vision of 6G:**

Transition from “Blind" to Environment-Aware communications

- **The Enabler**:

Radio Map constructs the Electromagnetic Digital Twin explicitly capturing spatial channel characteristics

- **Key Applications**
  - Pilot-free Channel Estimation
  - UAV Trajectory Planning
  - Proactive Beamforming



What Is a Radio Map and What Does "Sampling-Free" Mean?

Radio map as a spatial channel field

A radio map predicts channel quality over space.
It is conditioned on the environment, transmitter location, and frequency.
The output can be path loss, RSSl, SINR, AoA, or other spatial channel metrics.

$$\mathbf{M}:\{\mathbf{E},\mathbf{s},f\}\rightarrow\mathbf{H}(\mathbf{r})$$

E： environment layout and scene context

s ： transmitterposition

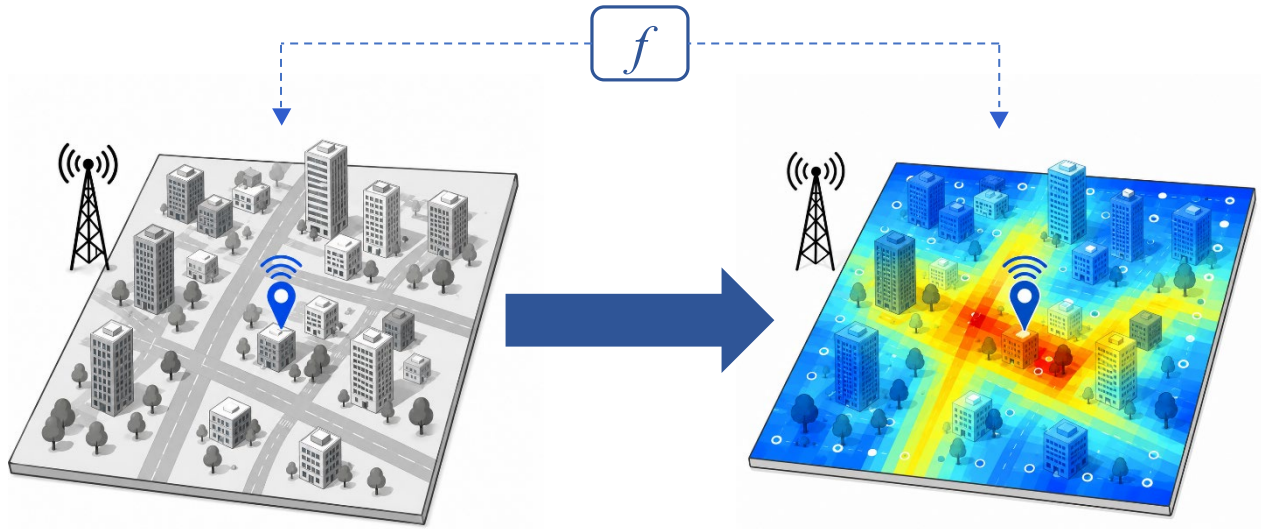
f ： carrier frequency

H(r)： channel value at location r

Key idea

The goal is to infer the full field from scene information,
Without exhaustive measurements.

From environment input to radio map output



Environment+transmitter **Predicted radiomap**

Important distinction

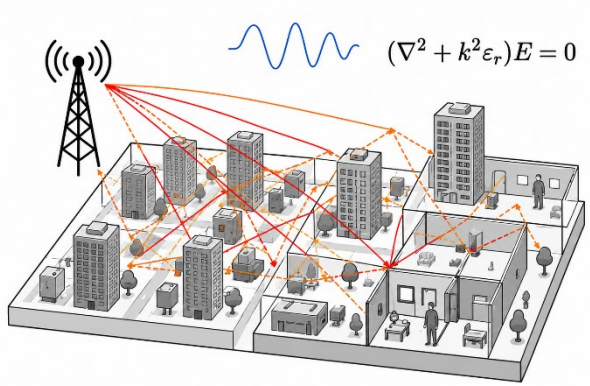
Sampling-free radio map construction: no in-situ channel measurements are required in the target area.

Diffusion sampling: iterative generative inference used inside a diffusion model.

Problem Statement: The Accuracy-Physics-Latency Trilemma

Three competing goals in radio map construction

Physics-based solvers



- High physical fidelity
- Very high computational cost
- Hard to update online

Full-wave



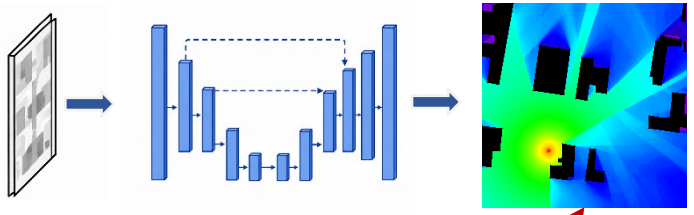
Ray tracing



Direct neural predictors

RadioUNet

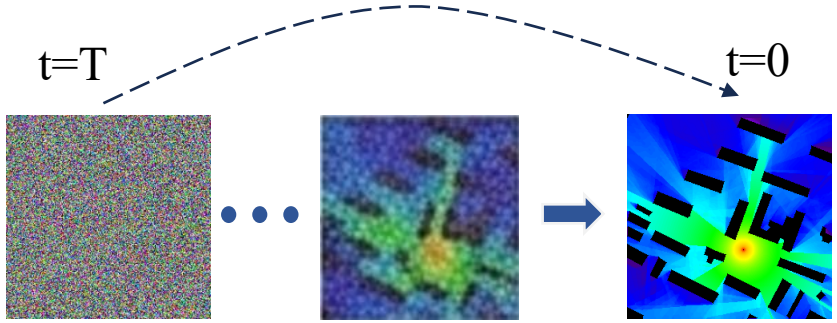
RME-GAN



Over-smoothed

- Millisecond inference
- Limited physical awareness
- Fine structures are blurred

Diffusion-based construction



- High reconstruction fidelity
- Strong detail recovery
- Iterative denoising is slower

Goal: High fidelity + physical consistency + low latency

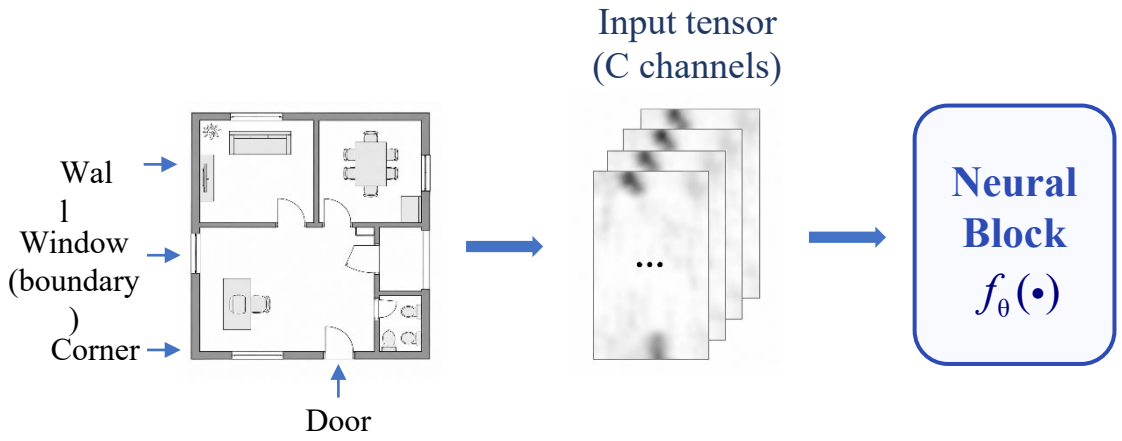
This tutorial explores three ways to inject physics into learning-based radio map construction.

Three Blind Spots of Diffusion-Based RM Construction

Why generic diffusion models still miss critical radio-map knowledge

Blind Spot 1

Where does physics act?

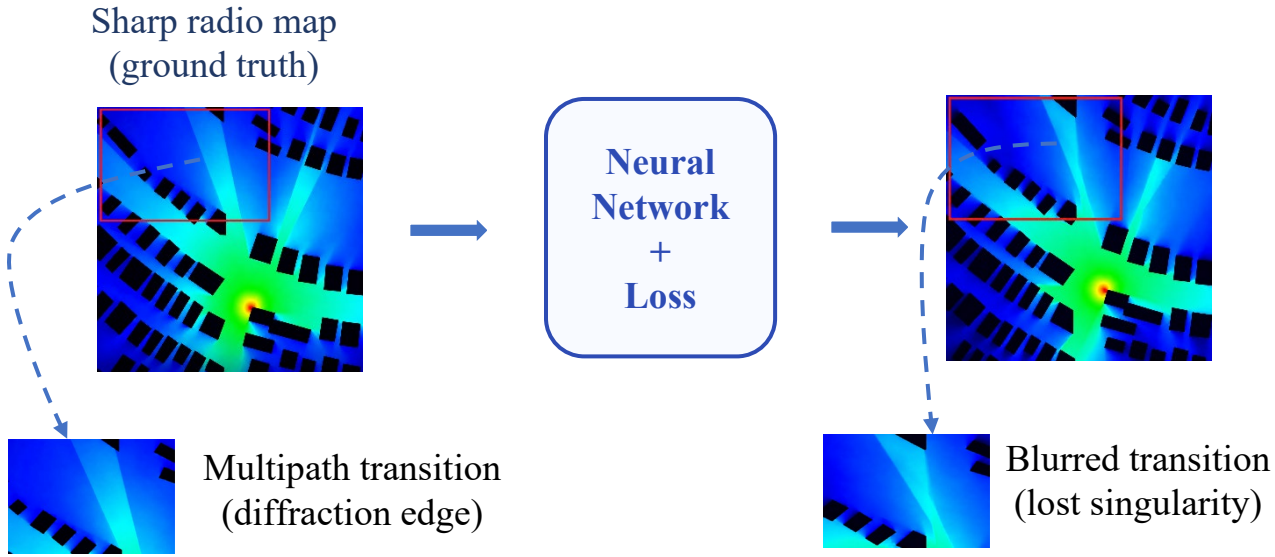


- Walls, doors, windows, and corners are often encoded as ordinary pixels.
- Material and boundary effects are weakly exposed.

iRadioDiff

Blind Spot 2

What field structures matter?



- Generic reconstruction losses do not emphasize multipath-induced singularities.
- Fine structural transitions are easily blurred.

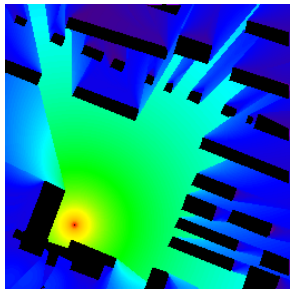
RadioDiff-k^2

X. Wang, N. Cheng, et. al. iRadioDiff: Physics-Informed Diffusion Model for Indoor Radio Map Construction and Localization. IEEE ICC 2026 Best Paper Award.
X. Wang, N. Cheng, et. al. RadioDiff-k^2: Helmholtz Equation Informed Generative Diffusion Model for Multi-Path Aware Radio Map Construction. IEEE JSAC 2026.

Three Blind Spots of Diffusion-Based RM Construction

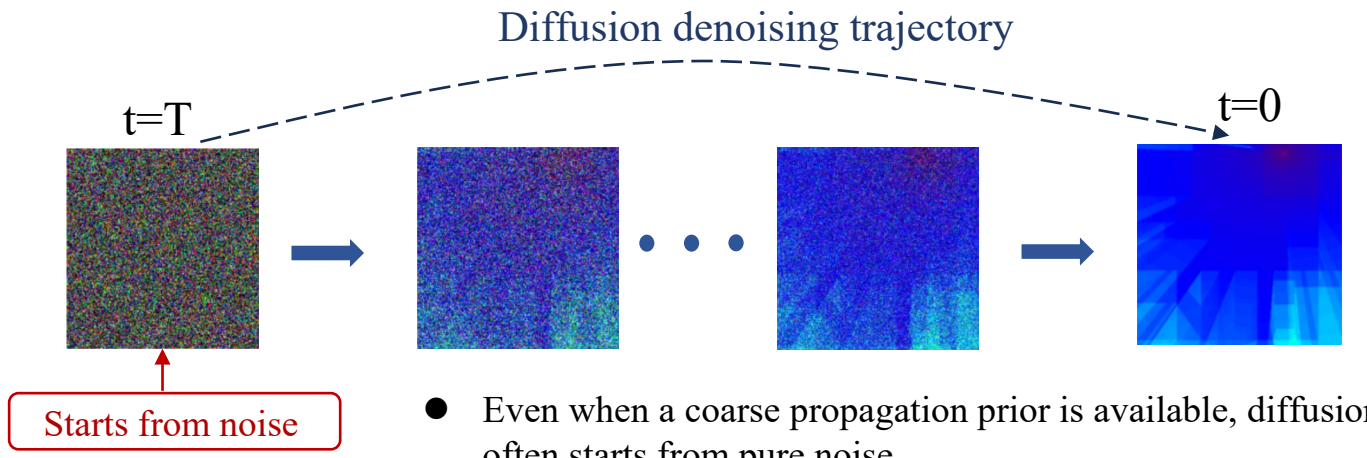
Why generic diffusion models still miss critical radio-map knowledge

Coarse propagation prior
(e.g., ray-based map)



Blind Spot 3

Where should sampling start?



- Even when a coarse propagation prior is available, diffusion often starts from pure noise.
- Early denoising steps may rebuild already known structure.

RMPrior

Three questions motivate this tutorial: condition the input, preserve the right structures, and shorten the reverse trajectory.

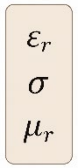
Each work injects physics at a different stage of radio map construction.

Tutorial Roadmap: Where Can Physics Enter the Pipeline?

Injection Point 1
Condition space
Geometry and material priors


Floor plan

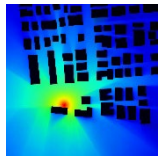

Wall/Window/Corner



Material tag


AP location

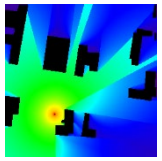
Injection Point 2
Representation space
Helmholtz-aligned singularity prior

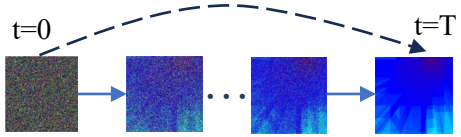

Sharp transition
(diffraction edge)


Outline/
Singularity map

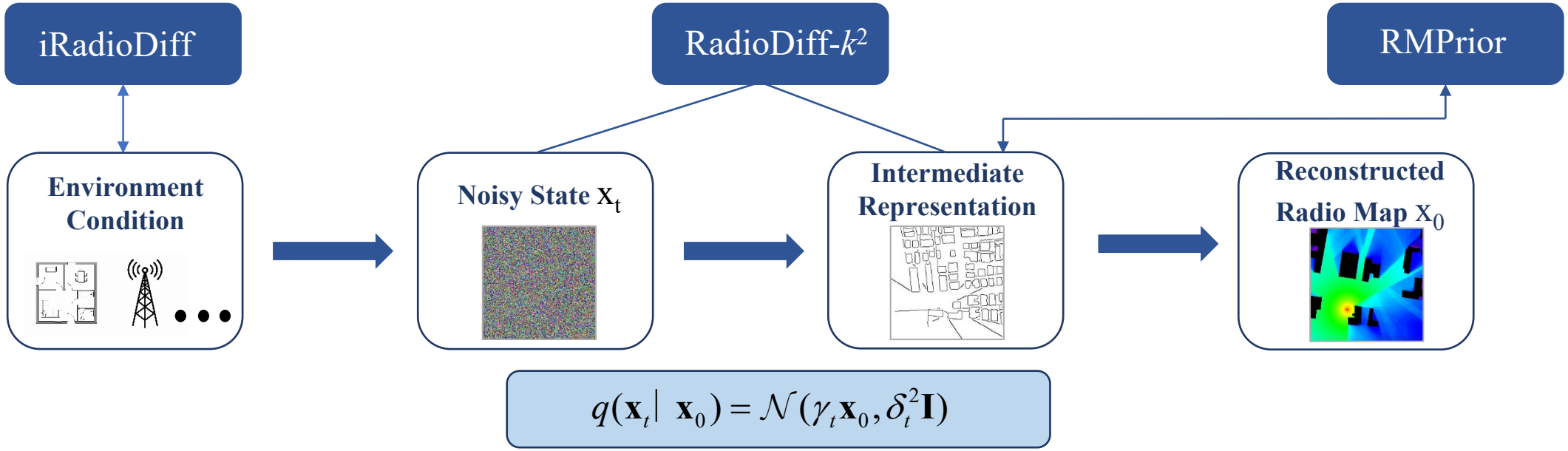
$$\nabla^2 u + k^2 u = 0$$
$$k_{\text{eff}}^2$$
Helmholtz prior
(local PDE)

Injection Point 3
Trajectory space
Propagation-informed initialization


Coarse propagation
prior(e.g., ray-based)



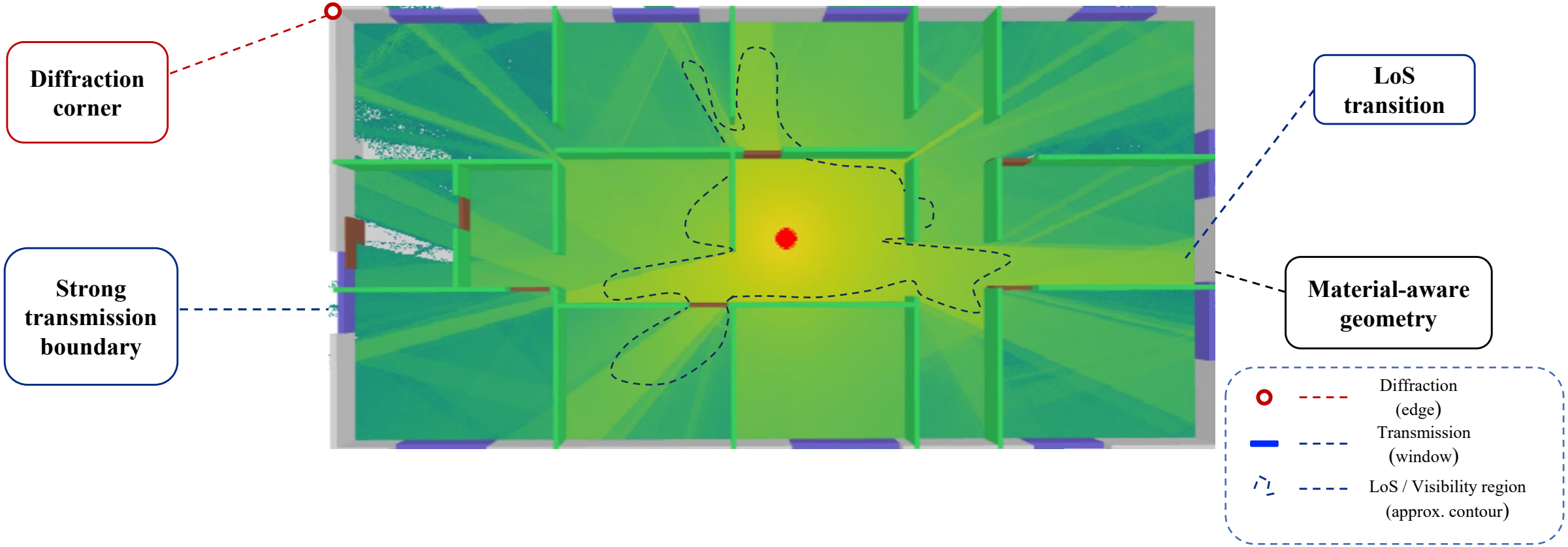
Denoising trajectory
(diffusion sampling)



This tutorial studies how physics can guide the input, the representation, and the inference trajectory.

Part I: Geometry- and Material-Aware Indoor Radio Maps

Where will the field change?



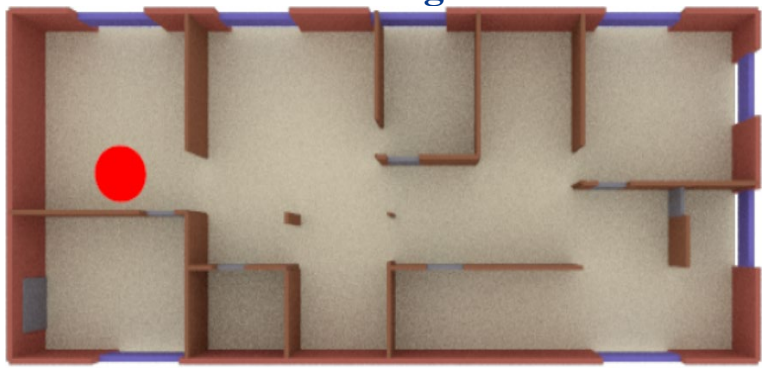
Physics acts at boundaries, corners, and material transitions.

Geometry prior **Material prior** **Boundary-aware learning**

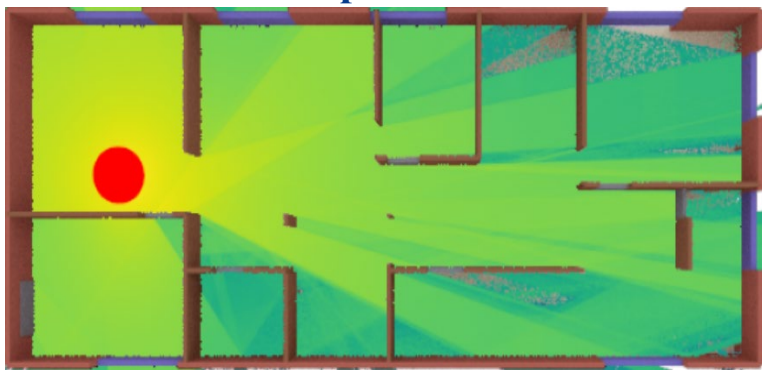
Indoor Propagation Is Not Determined by Geometry Alone

Same layout, different material configurations, different radio maps.

Material configuration A



RM output under A



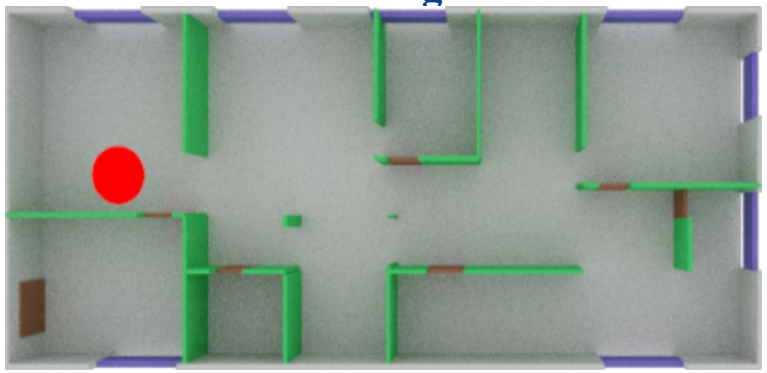
Material-dependent reflection

Partial transmission through doors and windows

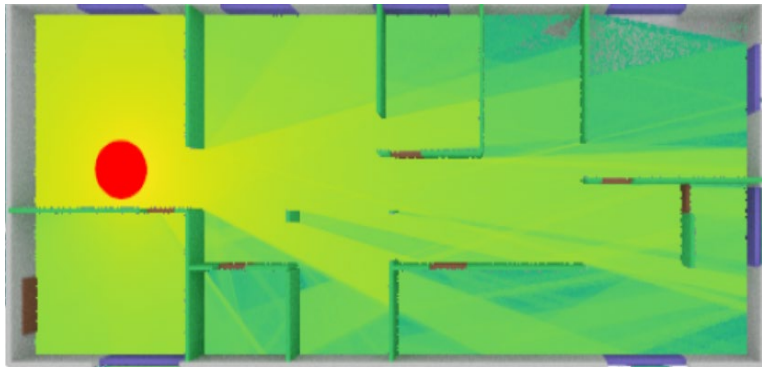
Diffraction near corners and wedges

Conceptual illustration

Material configuration B



RM output under B



A wall should be represented by both its shape and its electromagnetic interaction.

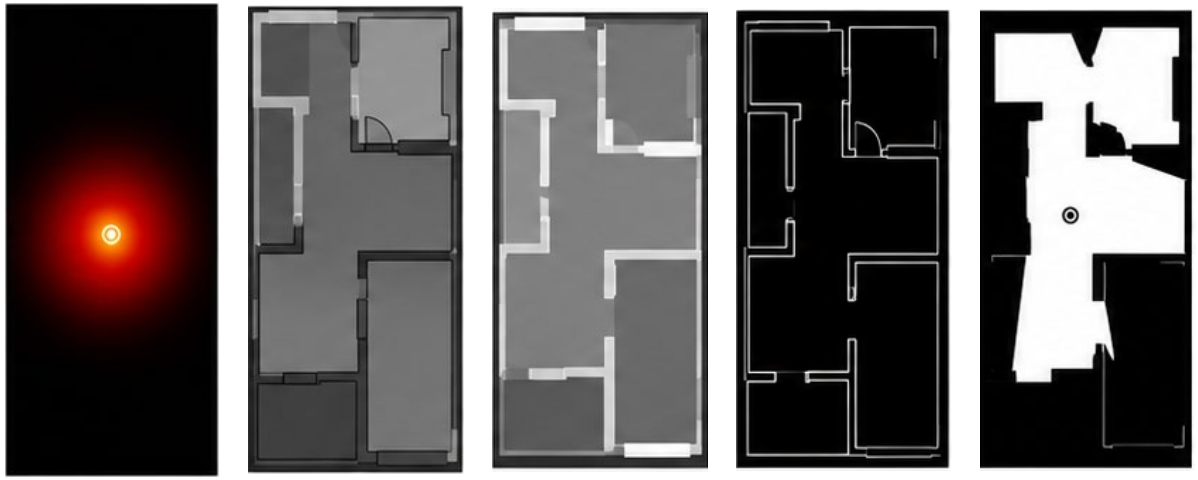
Same geometry | **Different materials** | **Different propagation fields**

From Floor Plans to Physics Prompts

Physics-Aware Condition



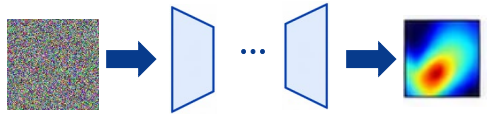
Environment + material annotation



AP location s | Reflection map H_r | Transmission map H_t | Boundary prior Γ | LoS mask

$$\mathbf{c} = \{s, H_r, H_t, \Gamma(s, C_e, B_t), LoS(s)\}$$

- **Material prompts:** how waves interact with surfaces
- **Boundary prompts:** where abrupt field changes may occur



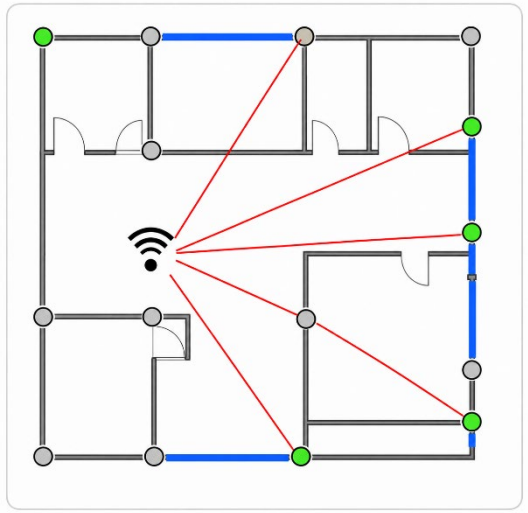
Noise x_t | Denoising (Diffusion / Generative Model) | Field map y

The condition is redesigned before the generative model is redesigned.

Where Do RSSI Discontinuities Occur?

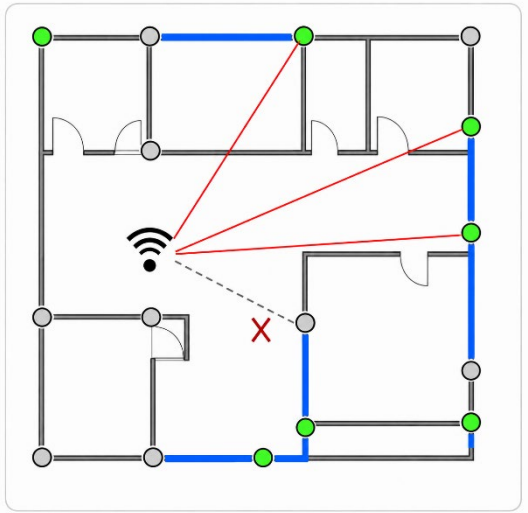
1. Detect candidates

Convex material corners and high-transmission boundaries



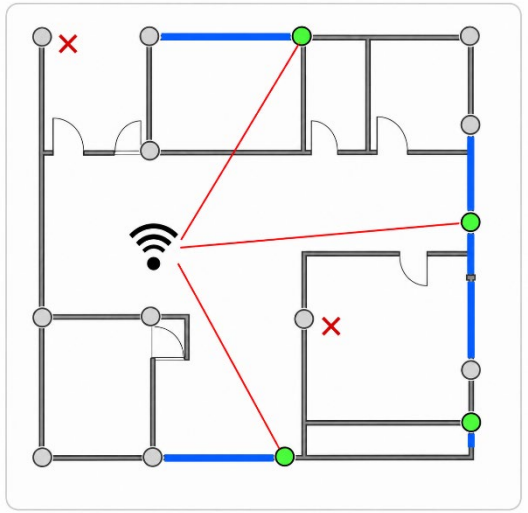
2. Connect boundary structures

Preserve wall-aligned contours, not trivial AP-to-corner rays



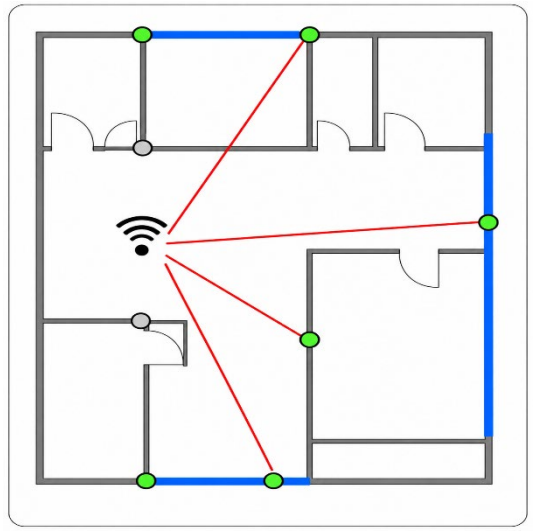
3. Apply physical pruning

No shadow, no diffraction



4. Add LoS–NLoS transitions

Form the final RSSI-discontinuity prior



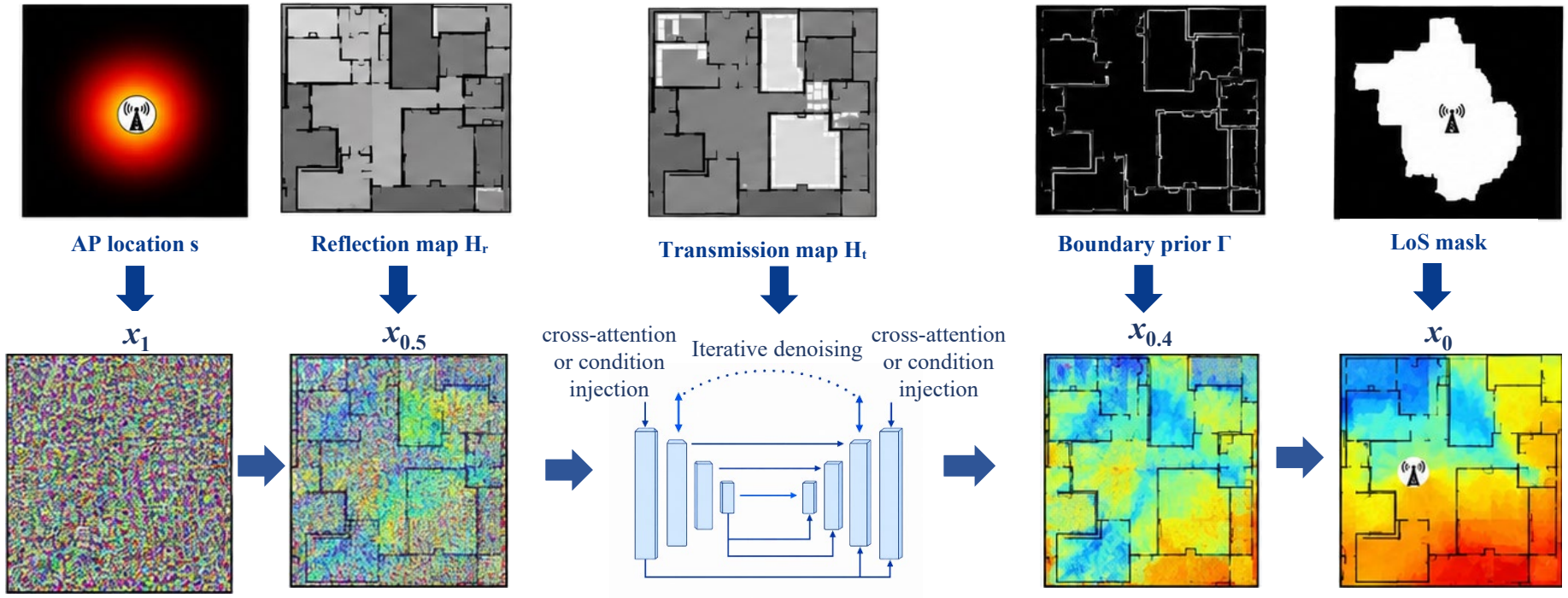
$$\Gamma(s, C_e, B_t) + LoS(s)$$

valid diffraction point | strong-transmission boundary | invalid candidate | AP-referenced ray

The model is told where continuity assumptions are most likely to fail.

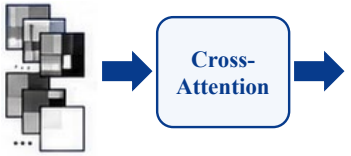
iRadioDiff: Physics-Prompted Conditional Diffusion

Physics-aware conditions



Conditional guidance

Spatial cross-attention adapts denoising to material and topology.



Boundary emphasis

Boundary-weighted learning highlights nonstationary field regions.

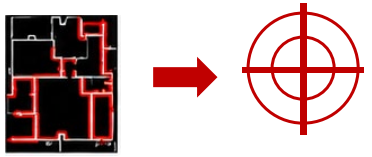


Image-space diffusion preserves fine-grained indoor RSSI structures.

Training: noisy image + ground-truth RM | **Inference:** iterative denoising from noise

Physics prompts guide both what the model sees and where it should learn more carefully.

Two Generalization Tests

Antenna Location Generalization (ALG)

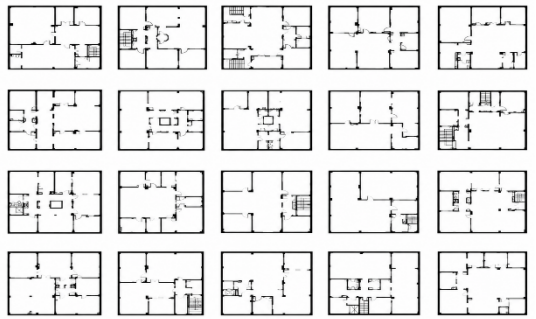


● = train AP (40) ● = test AP (10)

25 layouts observed during training → 40 of 50 AP positions used for training → 10 unseen AP positions used for testing

Zero-Shot Layout Generalization (ZLG)

Training layouts (20)



Unseen test layouts (5)

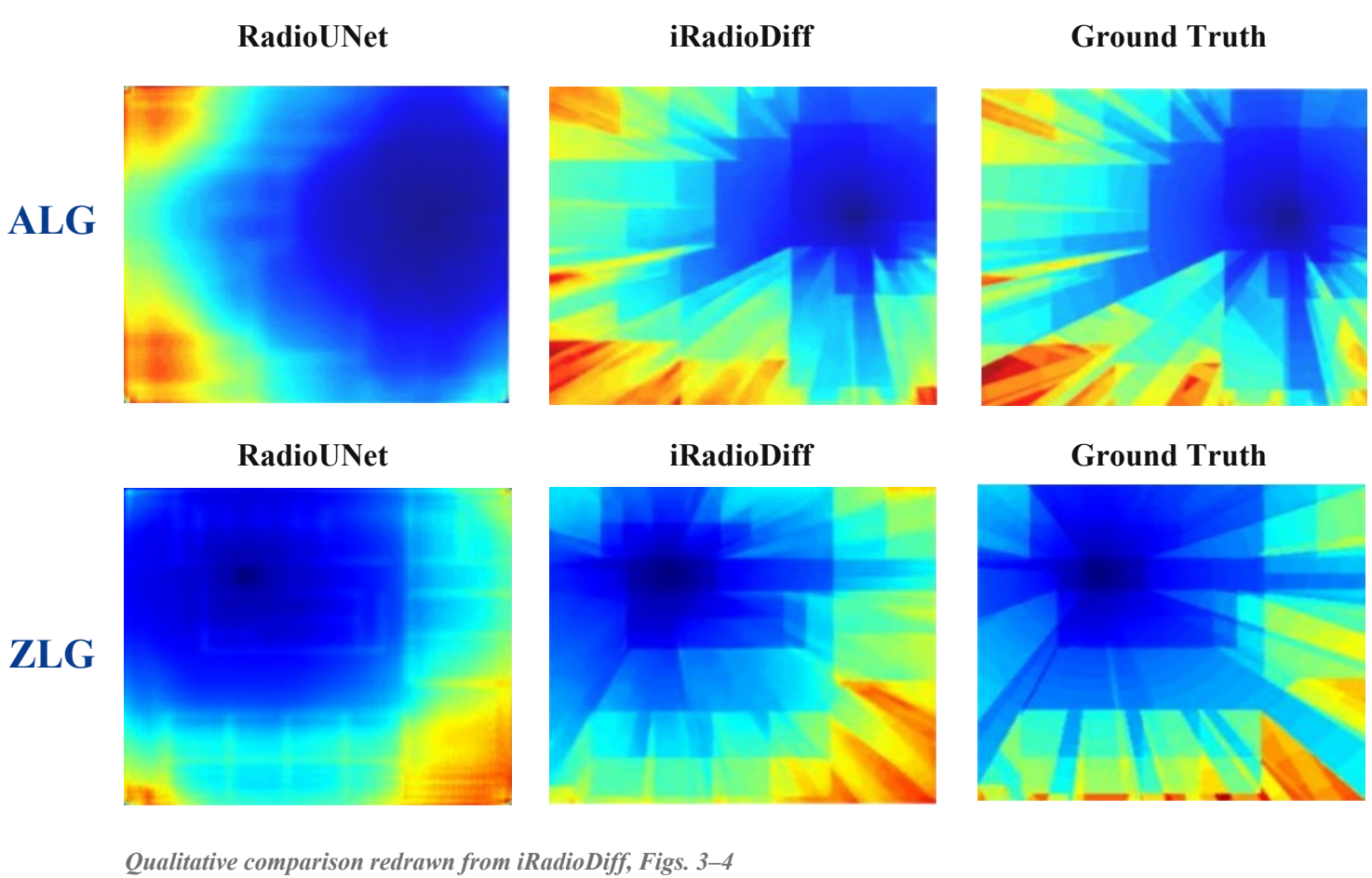


20 layouts used for training → 5 completely unseen layouts used for testing → 50 AP positions per test layout

3.5 GHz | 0.25 m resolution | 8 reflections | 10 transmissions | 2 diffractions | Tx/Rx height: 1.5 m

ALG tests deployment generalization; ZLG tests topology and material generalization.

iRadioDiff Results: Better Maps, Better Localization



Qualitative comparison redrawn from iRadioDiff, Figs. 3–4

ALG

RMSE: 6.357
PSNR: 32.24 dB
Localization error: 7.86 m

ZLG

RMSE: 7.010
PSNR: 31.45 dB
Localization error: 8.53 m

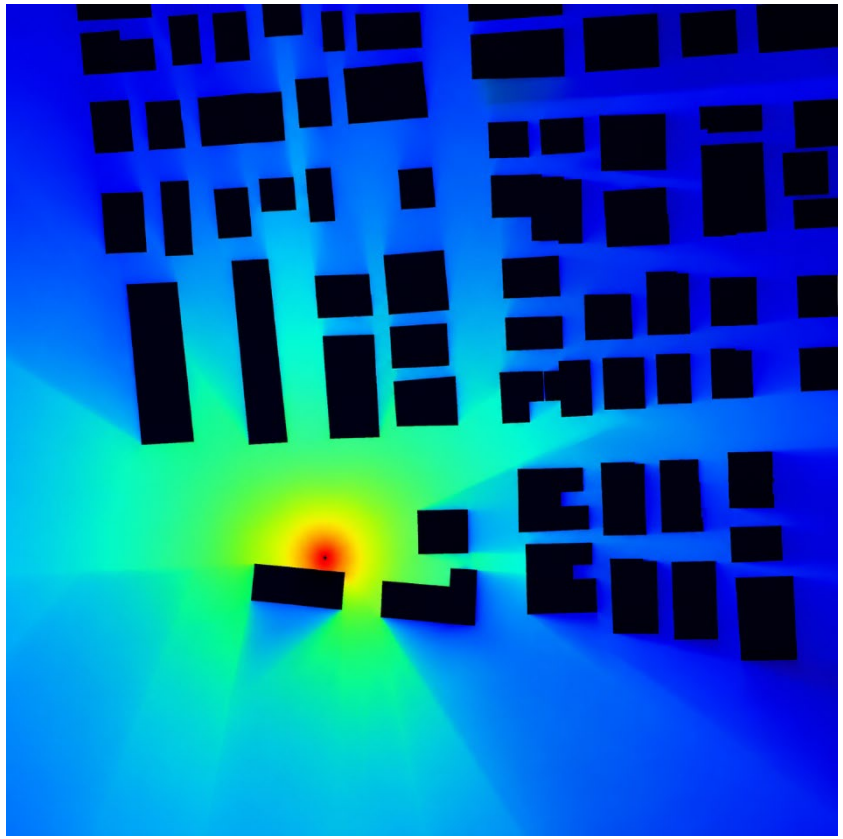
Physics-prior ablation

ALG	**RMSE:9.619** →	**6.357**
ZLG	**RMSE:11.75** →	**7.010**

In the ZLG localization ablation, the error changes from 8.765 m to 8.830 m. The reconstruction gains remain clear, but that specific localization ablation is not improved.

Physics prompts consistently improve RM fidelity and support strong RSSI-based localization in the full comparison.

Part II: Helmholtz-Aligned Priors for Multipath Singularities



What field structures must the model preserve?

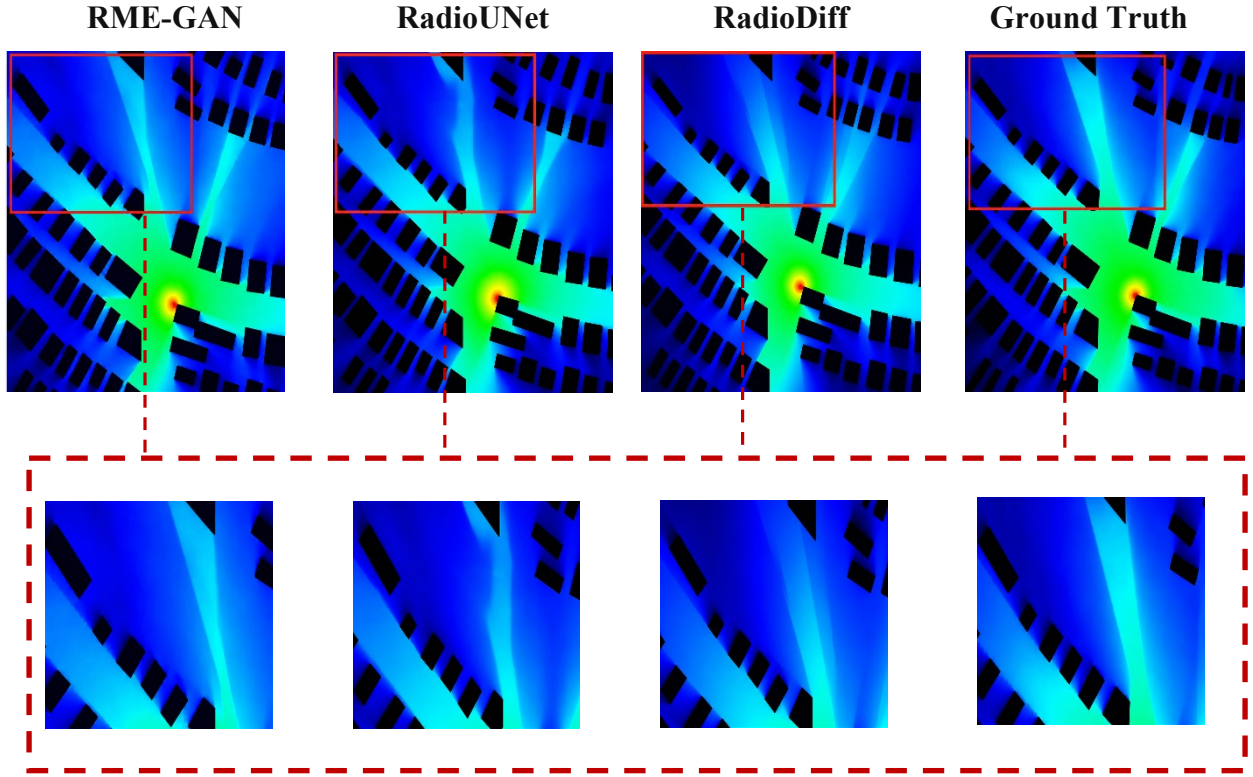
$$\nabla^2 u + k^2 u = 0$$

RadioDiff-k²



Image Edges Are Not EM Singularities

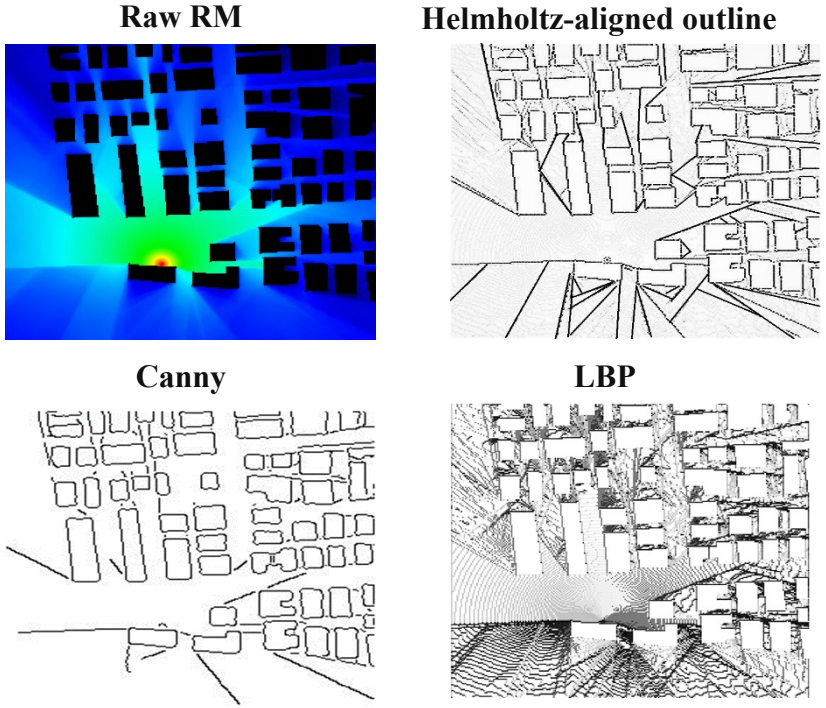
Missing High-Frequency Field Details



Oversmoothing removes abrupt path-loss transitions.

Visual concept redrawn from RadioDiff-k², Figs. 2–3.

Object Edges Are Not Field Singularities



Canny mainly follows geometric boundaries.
LBP produces dense local textures.
Helmholtz alignment targets field-transition structures.

A visually sharp edge is not necessarily a propagation-relevant singularity.

Starting Point: The Helmholtz Equation

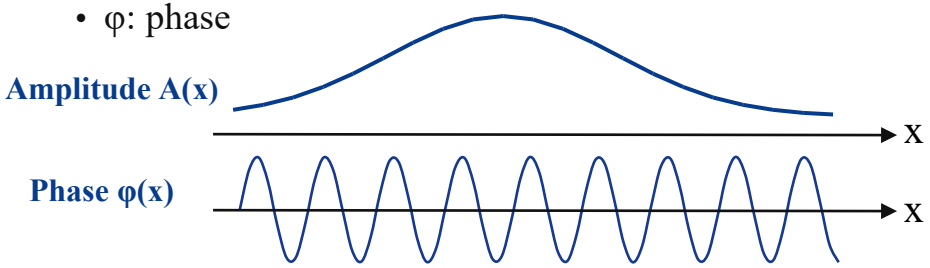
Locally isotropic, source-free, time-harmonic region

$$\nabla^2 u + k^2 u = 0$$

Complex field decomposition

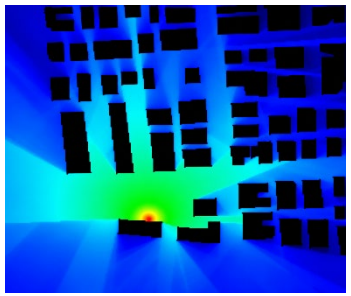
$$u(\mathbf{x}) = A(\mathbf{x}) e^{j\phi(\mathbf{x})}$$

- A: amplitude
- φ: phase

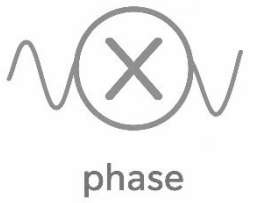


The field is complex-valued.

Observable radio-map supervision



$$I(\mathbf{x}) = |u(\mathbf{x})|^2 = A^2(\mathbf{x})$$







A radio map usually provides power, but not phase, complete boundaries, or material parameters.

The governing equation is complex-valued, while the available supervision is power-only.

From Complex Fields to Observable Amplitude Curvature

1 $$\nabla^2 A - A\|\nabla\varphi\|^2 + k^2 A = 0$$

Separate the real part after writing: $u = Ae^{j\phi}$

2 $$k_{\text{eff}}^2(x) \triangleq -\frac{\nabla^2 A(x)}{A(x)+\varepsilon}$$

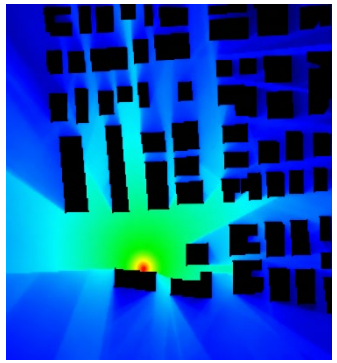
A local amplitude-curvature indicator computable from power observations.

3 $$k_{\log}(x) \triangleq -\nabla^2 \log(A(x)+\varepsilon)$$

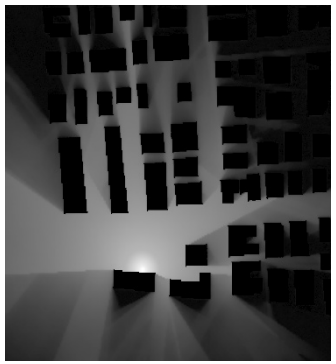
A gain-invariant alternative used to obtain a robust sign-based outline.

Observable proxy construction

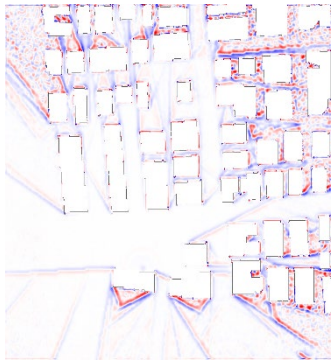
Power map I



Amplitude map



Signed amplitude-curvature map



Negative-region outline M

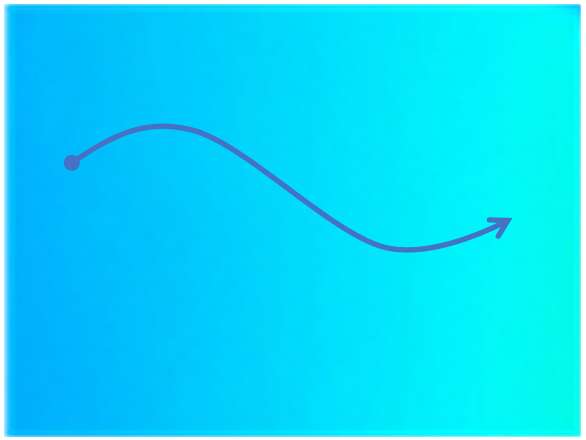


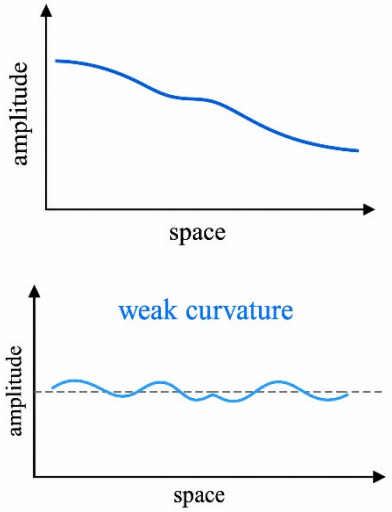
The PDE is translated into an observable intermediate representation, not imposed as a hard residual.

What Does a Negative Effective Indicator Mean?

Smooth Radiative Region

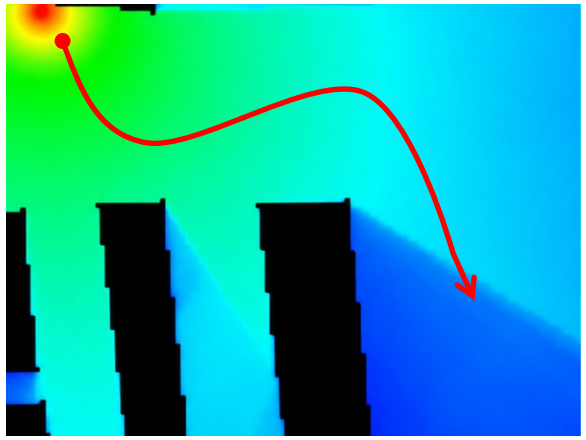
Gradual amplitude variation
Small local curvature
Stable spatial transition

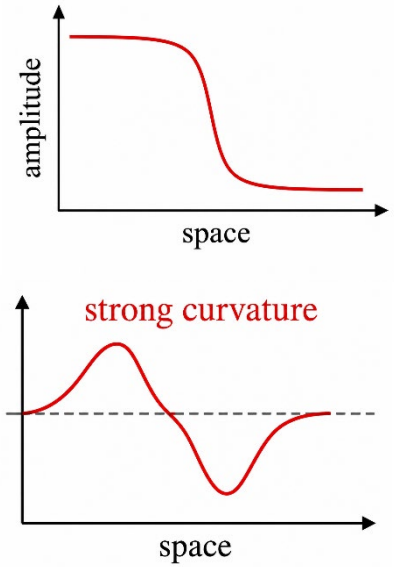




Shadowing, Diffraction, or Multipath Transition

Rapid amplitude variation
Strong local curvature
Negative indicator belt





$$k_{\text{eff}}^2(x) < 0$$

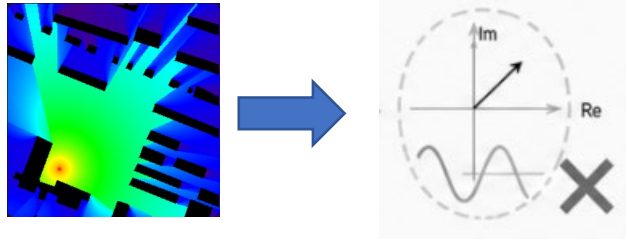
Important: k^2_{eff} is a data-derived curvature indicator. It is not the material wavenumber $\omega^2\mu\varepsilon$.

Negative regions are used as a physics-aligned indicator of abrupt field transitions.

Why Not Directly Enforce a PDE Residual?

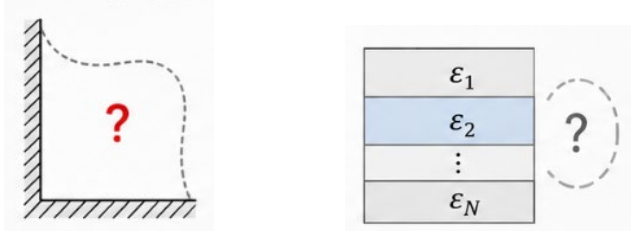
Missing Phase

RM measurements provide power, not the complex field.



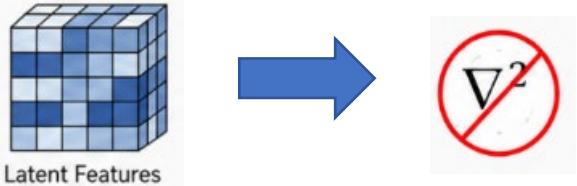
Uncertain Physics Parameters

Boundary conditions and material properties are often unavailable.



Nonphysical Latent Space

Differential operators have no direct physical meaning on latent features.



Observable Physics Representation

Compute the outline from training RMs, learn it with DM1, and use it to condition DM2.



Training RM

Outline (from RM)

Reconstructed RM

Use physics to define a measurable target before using physics to define a loss.

RadioDiff-k²: Structure First, Field Second

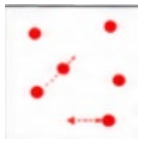
Offline Training

Environment, BS, and Ground-Truth RM

Static map H_s



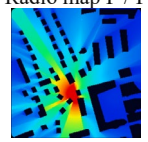
Dynamic map H_d



BS / Transmitter



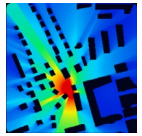
Radio map P / I



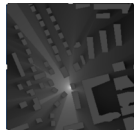
$\{H_s, H_d, r\}, P$

Stage 0: Physics-label generation

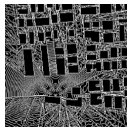
Power map I



Amplitude $A = \sqrt{I}$



Effective curvature k_eff^2



Outline map M



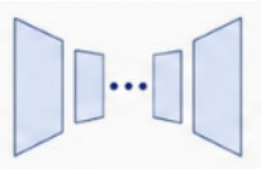
$I \rightarrow A \rightarrow k_eff^2 \rightarrow M$

Helmholtz-aligned outline label



M

Stage 1: Singularity generation

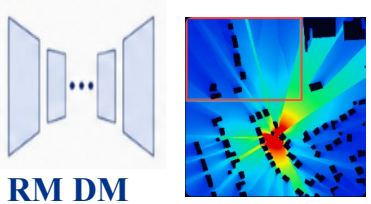


Outline DM

$\{H_s, H_d, r\} \rightarrow M^{\wedge}$

Learned diffusion stage (DM1)
Trained to predict the outline

Stage 2: Radio-map generation



RM DM

$\{H_s, H_d, r, M^{\wedge}\} \rightarrow P^{\wedge}$

Learned diffusion stage (DM2)
Trained to generate the radio map

Structure first, field second.

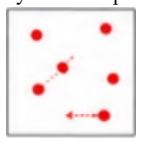
Online Inference

Environment and BS

Static map H_s



Dynamic map H_d

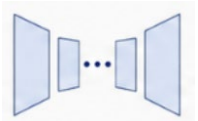


BS / Transmitter



$\{H_s, H_d, r\}$

DM1



Outline DM

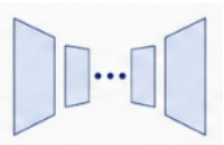
$\{H_s, H_d, r\} \rightarrow M^{\wedge}$

Predicted outline



$M^{\wedge}$

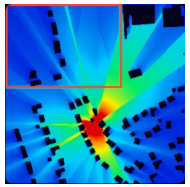
DM2



RM DM

$\{H_s, H_d, r, M^{\wedge}\} \rightarrow P^{\wedge}$

Reconstructed RM



$P^{\wedge}$

Two learned diffusion stages

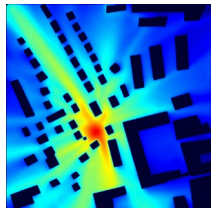
Helmholtz preprocessing is used only to create training labels. It is not executed during online inference.

The first diffusion model reduces the structural burden faced by the second.

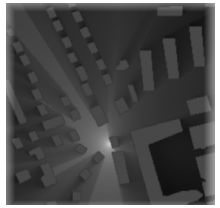
Training Uses Physics Labels; Inference Uses Neural Prediction

OFFLINE

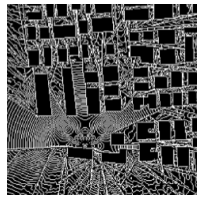
Physics-derived label generation for training supervision



Ground-Truth Radio Map



Field Amplitude $A = \sqrt{I}$



Finite-Difference Curvature



Outline Supervision

$$L_total = \lambda_1 L_drift + \lambda_2 L_noise + \lambda_3 L_recon$$

Latent DDM Backbone

VAE-Compressed Representation

No Online Helmholtz Preprocessing

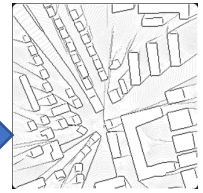
ONLINE

Neural prediction without physics preprocessing



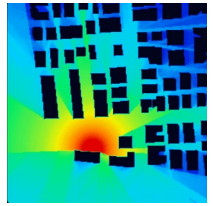
Environment and BS

DM1-Predicted Outline



Predicted Outline

DM2-Generated Radio Map



Reconstructed Radio Map

Physics provides offline supervision for the intermediate outline target.
Online prediction is performed by two learned diffusion stages.

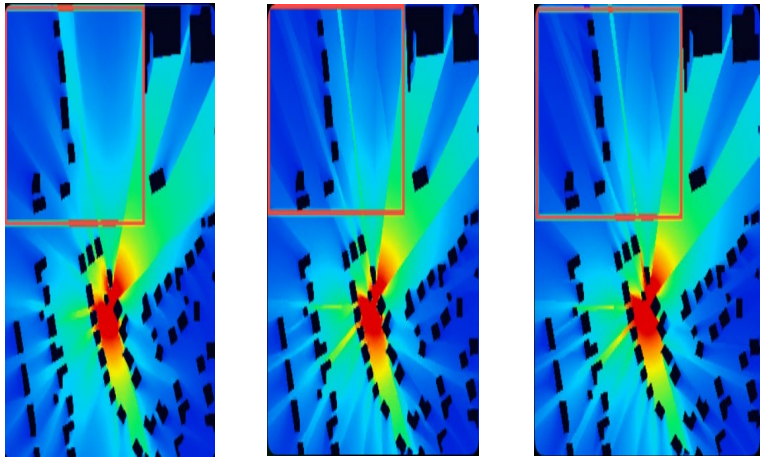
Physics supervises the intermediate target; neural generators perform online prediction.

Results Across Static, Dynamic, and Multipath Maps

Relative NMSE gains are measured against RadioDiff within each dataset.

Static Radio Map (SRM)

RadioDiff | RadioDiff-k² | Ground Truth



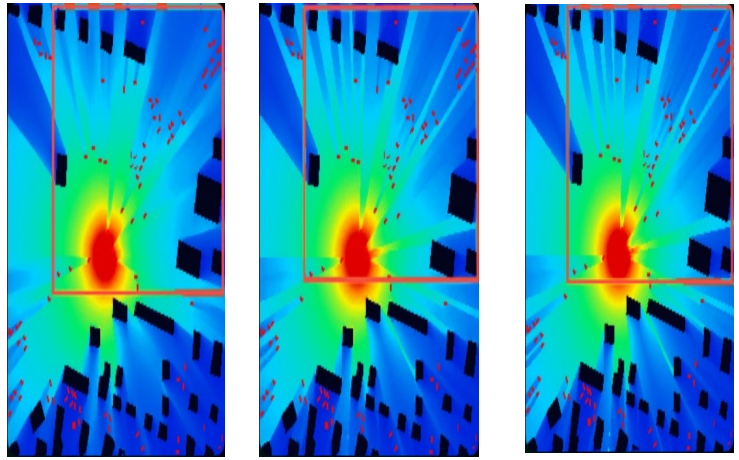
NMSE: 0.0043

PSNR: 34.46 dB

40.28% lower NMSE than RadioDiff

Dynamic Radio Map (DRM)

RadioDiff | RadioDiff-k² | Ground Truth



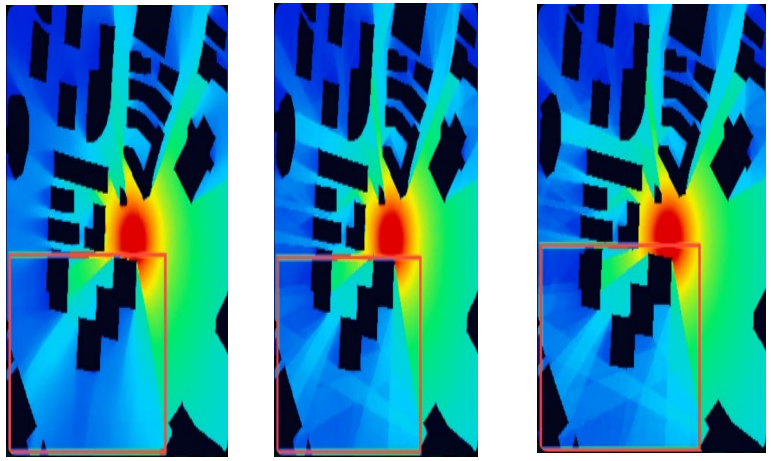
NMSE: 0.0054

PSNR: 33.79 dB

40.00% lower NMSE than RadioDiff

Multipath-Aware Radio Map (MRM)

RadioDiff | RadioDiff-k² | Ground Truth



NMSE: 0.0066

PSNR: 32.68 dB

45.45% lower NMSE than RadioDiff

The largest relative NMSE gain appears in the multipath-aware setting.

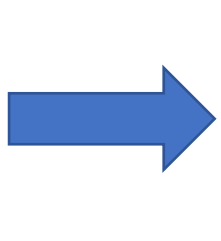
Ablation: The Prior Must Be Physics-Specific

Generic Texture Priors Can Hurt

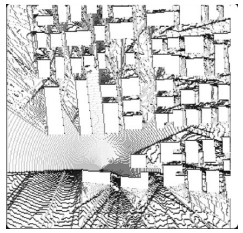
Prior (Condition) | Resulting MRM Radio Map (dB)

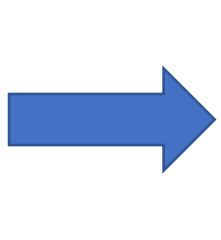
Canny outline



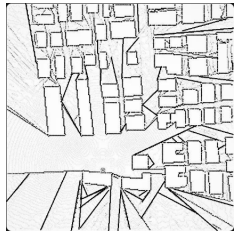


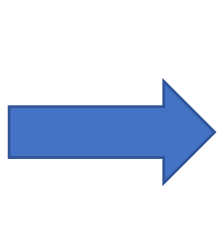
LBP texture





Helmholtz-aligned outline





The same MRM crop is used for all three comparisons.

MRM PSNR

Canny:	**11.97 dB**
LBP:	**12.06 dB**
RadioDiff:	**30.44 dB**
RadioDiff-k²:	**32.68 dB**

Outline Prediction, DTC

SRM: U-Net 0.8141 → **Diffusion 0.9703**

DRM: U-Net 0.6857 → **Diffusion 0.9782**

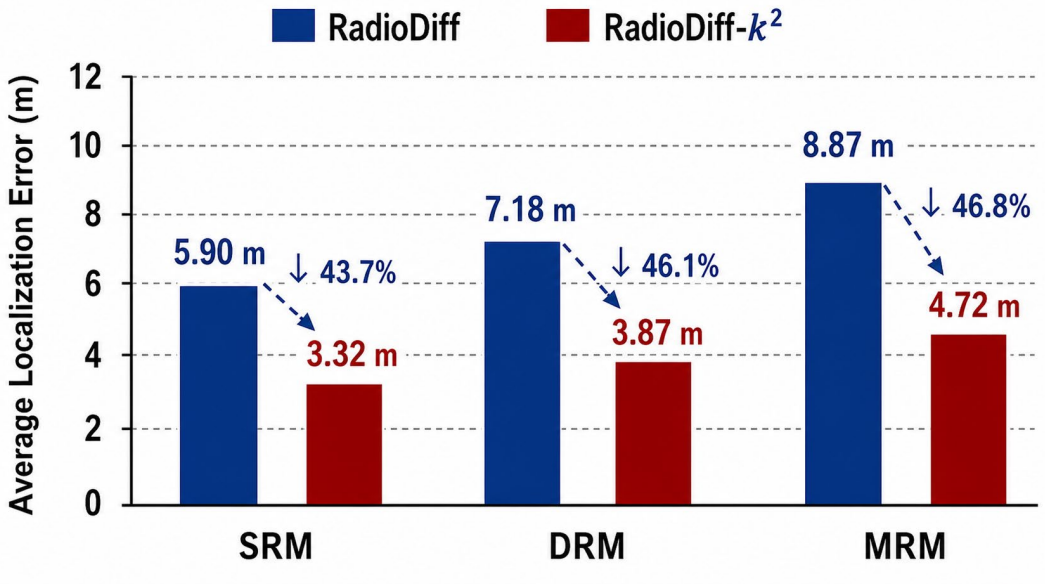
MRM: U-Net 0.7433 → **Diffusion 0.9577**

Diffusion predicts fine-grained outlines more accurately than U-Net.

An additional condition is useful only when it encodes task-relevant physics.

Practical Value and Computational Cost

Localization Error with KNN, K = 5



Sub-5-meter localization is achieved by RadioDiff-k² across all three settings.

Inference Cost

Model	Time	Memory
RadioDiff	0.24 s	1.52 GB
RMDM	19.30 s	1.65 GB
RadioDiff-k²	**0.76 s**	**3.02 GB**

Two diffusion stages improve structure but increase latency and memory over RadioDiff.

Time vs RadioDiff: **3.17×**

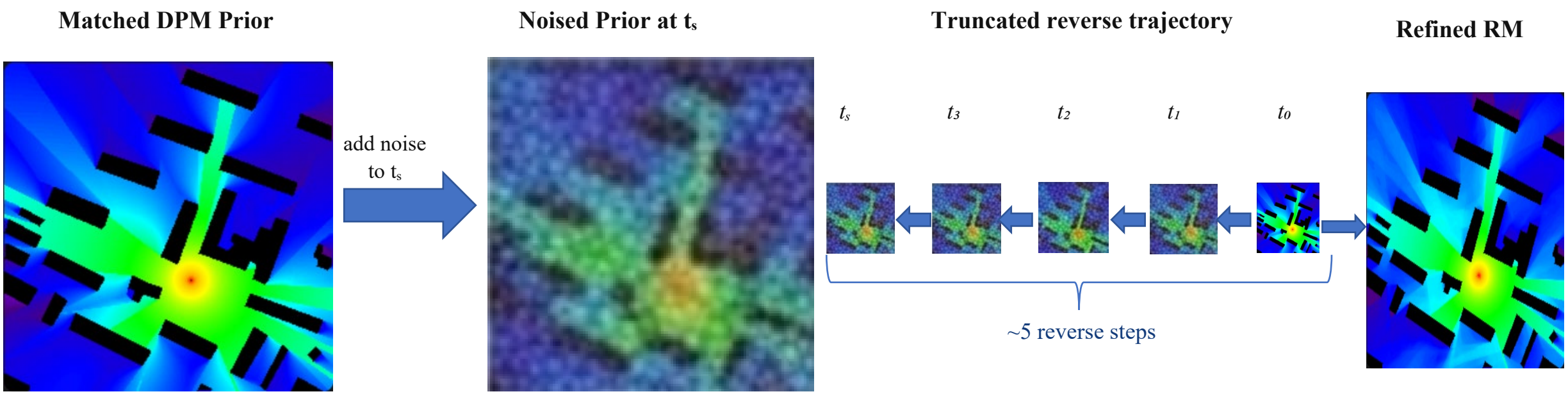
Memory vs RadioDiff: **1.99×**

Physics-guided decomposition improves accuracy and utility, but does not solve iterative-generation cost.

Part III: Propagation-Prior Guided Diffusion Acceleration

Where should denoising start?

RMPrior

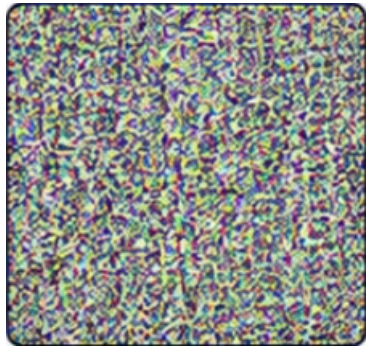


Physics / Channel Knowledge → **Matched Prior** → **Shorter Reverse Chain, Better Efficiency**

Z. Guo, X. Wang, N. Cheng. RMPrior: Bridging Propagation Priors and Diffusion Refinement for Efficient Radio Map Construction. IEEE GlobeCom 2026. (Online available)

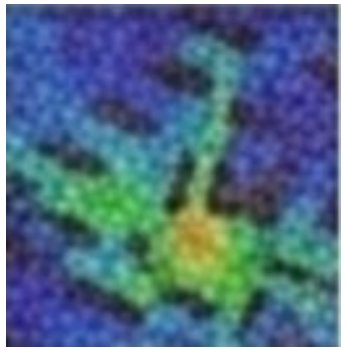
The Hidden Waste in Full Diffusion Sampling

Standard Full-Chain Sampling



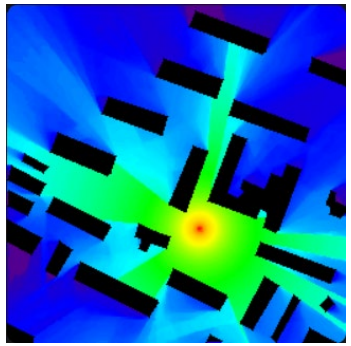
Gaussian noise

Rebuild scene layout



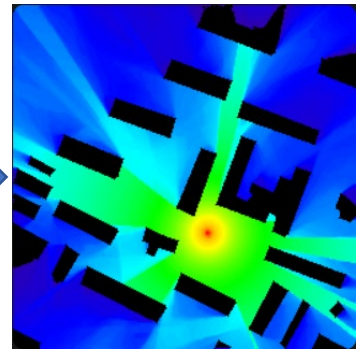
global propagation layout

Recover dominant attenuation



local attenuation details

Refine multipath details



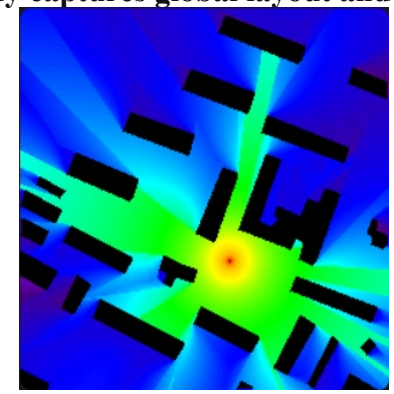
RM

?

Redundant reconstruction when a matched prior is available

Available Propagation Knowledge

DPM prior already captures global layout and dominant blocked regions







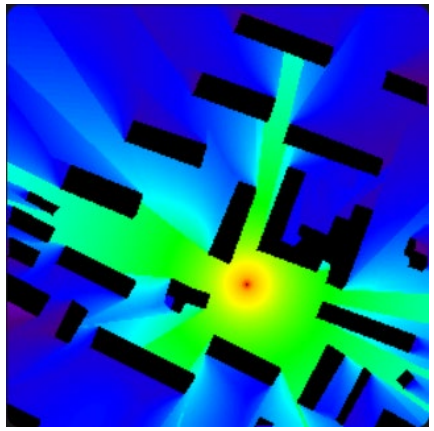
The diffusion model should refine missing multipath details, not rebuild the entire scene from noise.

RMPrior: Start from a Noised Propagation Prior

$$x^{p}_{ts} = \sqrt{(\bar{\alpha}_{ts})}\, x_p + \sqrt{(1 - \bar{\alpha}_{ts})}\, \varepsilon, \qquad \varepsilon \sim N(0, I)$$

$$t_s = [P_start\ T]$$

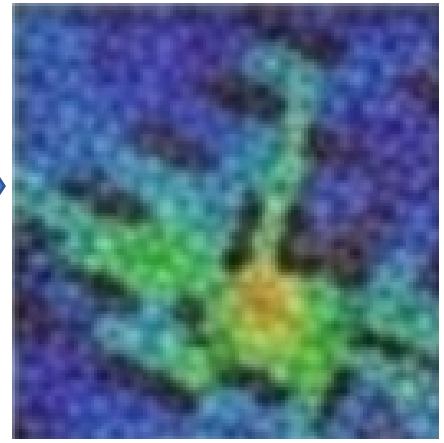
Matched propagation prior x_p



Forward noising

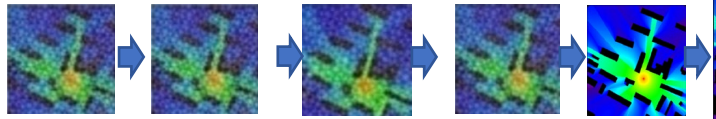


Intermediate state x^{p}_{ts}

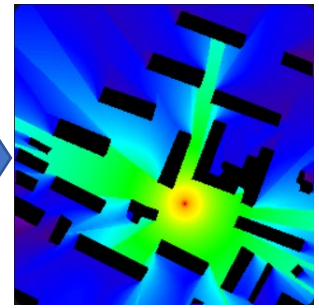


$t_s \rightarrow t_s - 1 \rightarrow \ldots \rightarrow 1 \rightarrow 0$





Reconstructed RM



$T \longrightarrow t_s \longrightarrow 1 \rightarrow 0$

(Removed early steps) (Active truncated reverse chain)

Inference-only modification **No backbone retraining** **Scene-aligned initialization**

The prior supplies the global layout; diffusion performs residual multipath refinement.

P_{start} Splits the Work Between Physics and Learning

$P_{start} = 1.0$

Full reverse chain

Weak dependence on the prior

RECOMMENDED

$P_{start} = 0.5$

Balanced refinement

50 reverse evaluations when T = 100

$P_{start} = 0.25$

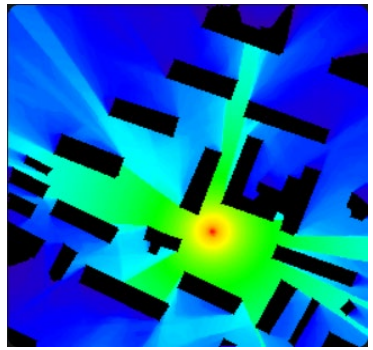
Aggressive truncation

Strong dependence on prior quality

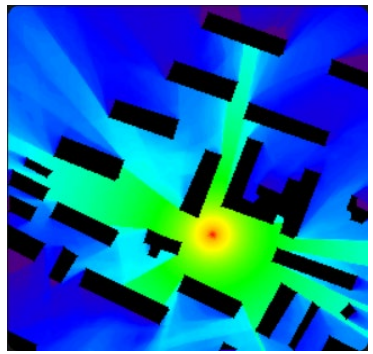
1.0 **0.25**

P_{start} (decreasing)

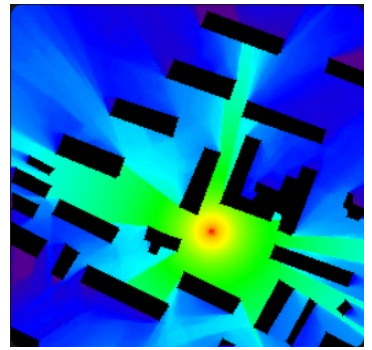
P_{start} controls both computation and dependence on prior quality.



1.0



0.5

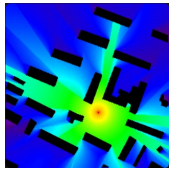


0.25

$$N_FE \approx P_{start}\ T$$

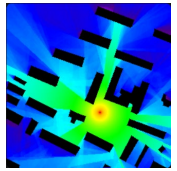
Smaller P_{start} saves computation but leaves less capacity to correct prior mismatch.

Theory: Prior Mismatch vs Removed Denoising Error



Prior x_p

vs.

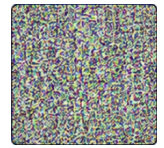


Target x_0

Initialization Gap

$$\|x^{p}_{ts} - x^{GT}_{ts}\|_2 = \sqrt{\bar{\alpha}_{ts}}\ \|x_p - x_0\|_2 \ .$$

The intermediate-state discrepancy is the prior mismatch scaled by the forward noising factor.



Noised state x_{ts}

Contracted prior mismatch

$$\sqrt{\bar{\alpha}_{ts}}\ \|x_p - x_0\|_2$$





Accumulated denoiser error removed by truncation

$$\Sigma^{T}_{t=ts+1}\ \delta_t\ \Pi^{t-1}_{j=1}\ \gamma_j$$

1 2 ... t_s t_s+1 ... T

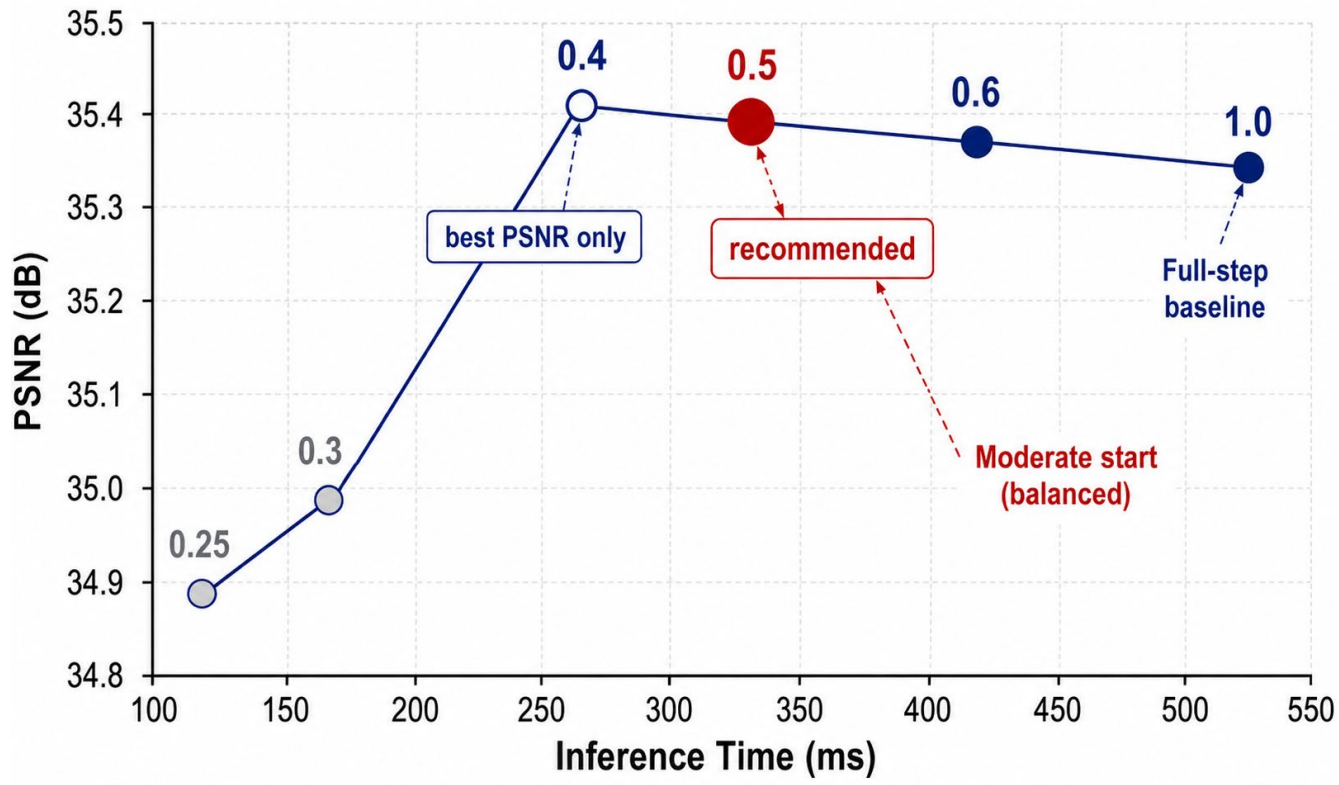
$$\sqrt{\bar{\alpha}_{ts}}\ \|x_p - x_0\|_2\ \Pi^{t}_{st=1}\ \gamma_t \ < \ \Sigma^{T}_{t=ts+1}\ \delta_t\ \Pi^{t-1}_{j=1}\ \gamma_j$$

prior gap after contraction — error removed from early steps

The sufficient condition assumes a contractive reverse process with per-step approximation error.

Moderate truncation can improve fidelity when removed early-step error exceeds the propagated initialization gap.

Results: A Better Speed-Fidelity Frontier



Recommended Operating Point:

$P_{start} = 0.5$

514 ms → **256 ms**

2.01× speedup

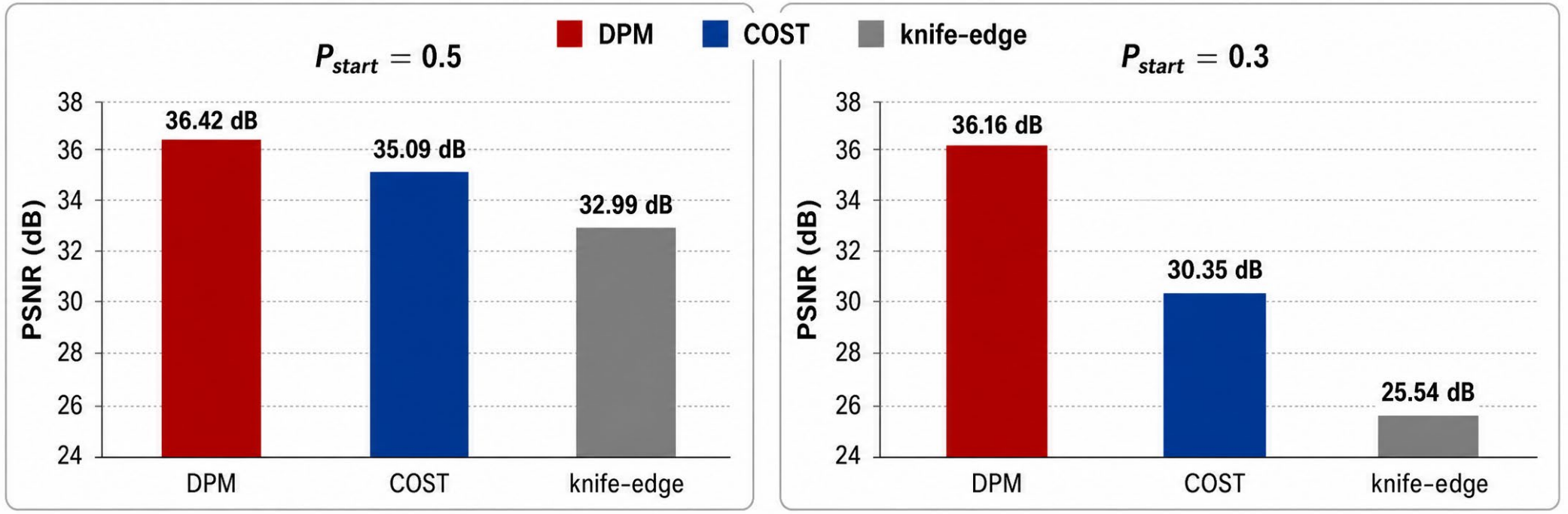
NMSE:	0.00688	→	**0.00683**
RMSE:	0.01808	→	**0.01794**
SSIM:	0.93910	→	**0.93935**
PSNR:	35.28	→	**35.39 dB**

P_{start} = 0.4 reaches 35.41 dB PSNR, but its NMSE, RMSE, and SSIM are worse than those at 0.5.

A moderate start improves both latency and aggregate reconstruction fidelity.

Prior Quality Determines How Aggressive We Can Be

Fewer correction steps



DPM-COST gap:
1.33 dB → **5.81 dB**

DPM-knife-edge gap:
3.43 dB → **10.62 dB**

Runtime is nearly unchanged across prior types at a fixed P_{start}.

Aggressive truncation amplifies prior-quality differences.

Deployment View: When Is Mid-Start Truly Faster?

End-to-end benefit decreases

Cached Prior

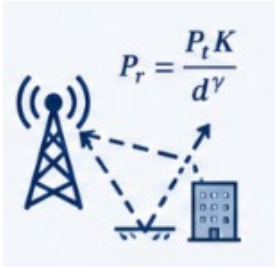


The same scene prior is reused across repeated updates.

Strongest end-to-end benefit

Best case

Lightweight Analytical Prior



Prior generation is much cheaper than the removed denoising steps.

Clear net acceleration

Net positive

Expensive Per-Query Prior



Prior generation may dominate the total latency.

Diffusion speedup does not equal system speedup

Caution

$$T_total = T_prior + T_truncated\ denoising$$

The paper reports the denoising latency on an RTX 4090. Prior-generation cost is deployment-dependent.

RMPrior is most attractive when a matched prior is cheap, cached, or reusable.

Three Papers, One Physics-Informed Design Space

Where will propagation change?

What structures must be preserved?

How far should diffusion travel?

Method	Knowledge Source	Injection Point	Failure Mode Addressed	Main Benefit	Main Cost or Risk
iRadioDiff	Geometry and material properties	Condition and boundary emphasis	Indoor material heterogeneity and discontinuities	Generalization and localization	Geometry preprocessing and annotation requirements
RadioDiff-k²	Helmholtz-aligned amplitude curvature	Intermediate outline	Multipath singularity oversmoothing	Fidelity and downstream utility	Two diffusion stages
RMPrior	Matched propagation model	Sampling initialization	Redundant early denoising	Inference acceleration	Sensitivity to prior mismatch

Physics is most useful when it targets a specific failure mode at a specific stage.

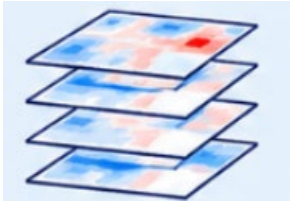
A Practical Recipe for Choosing the Right Prior

What knowledge is available?

Layout and material labels

Condition-space prior

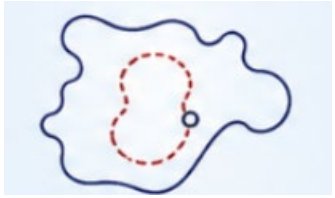
iRadioDiff-like design



High-fidelity training RMs and a governing equation

Representation-space prior

RadioDiff-k²-like design



Fast matched propagation model at inference

Trajectory-space prior

RMPrior-like design



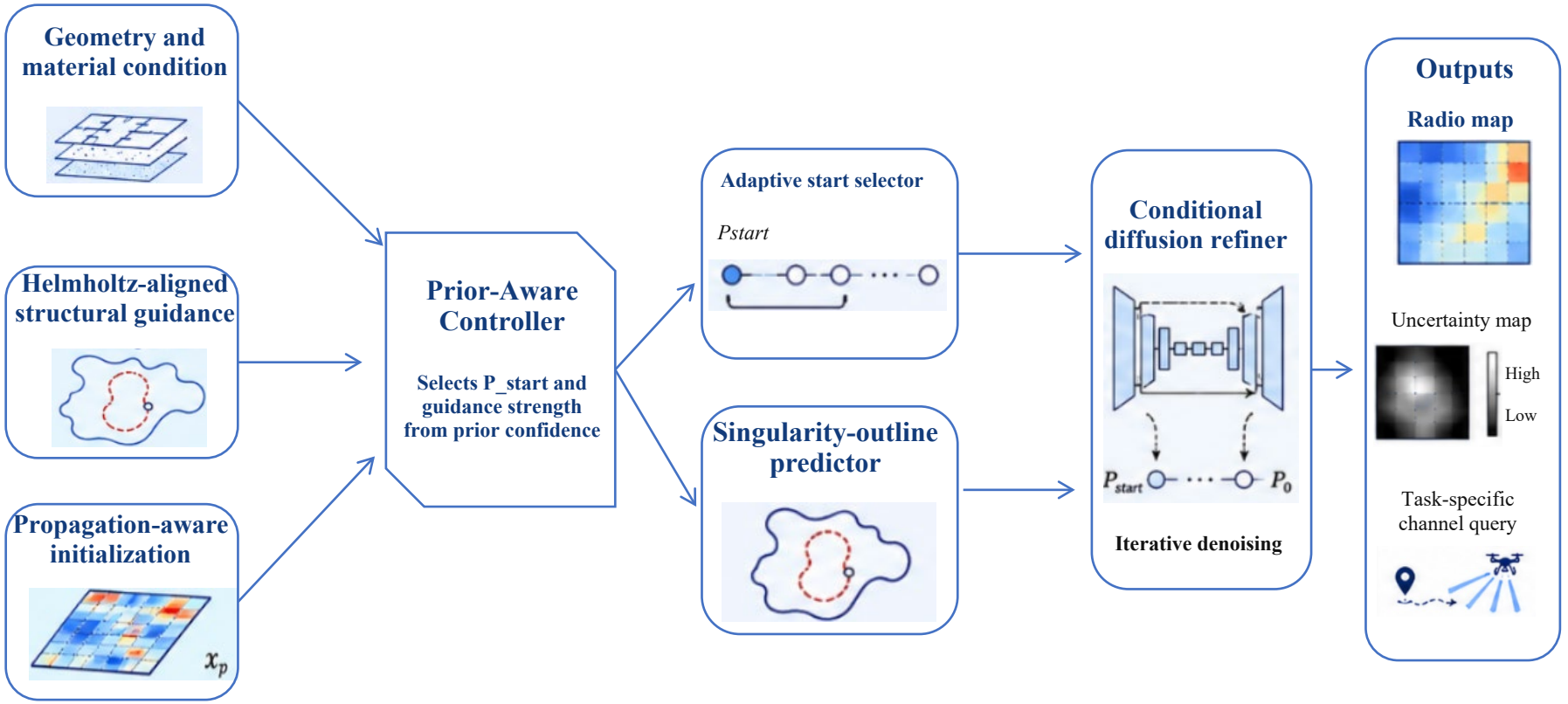
What must be validated?

- ☐ Physics-specific ablation
- ☐ Generalization to unseen layouts or transmitters
- ☐ Downstream communication utility
- ☐ End-to-end latency
- ☐ Sensitivity to prior mismatch

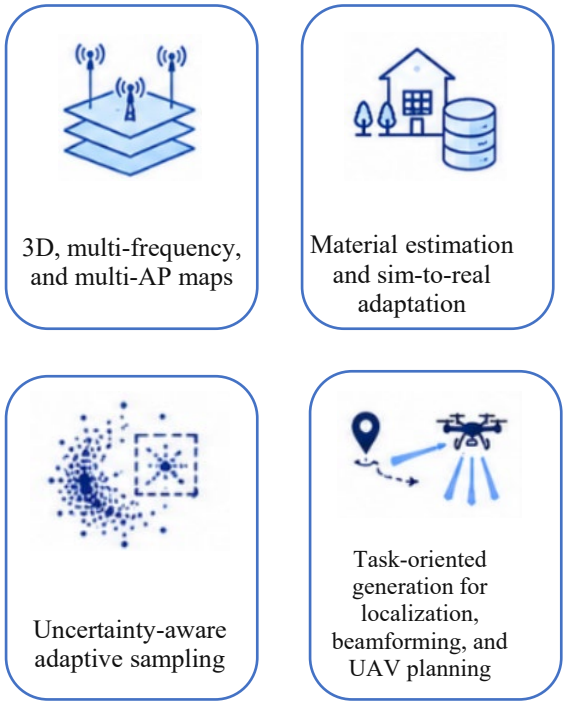
Select the prior according to available knowledge, then test the failure mode it is meant to solve.

Toward a Unified Physics-Informed RM Generator

A Unified Generator Could Combine All Three Priors



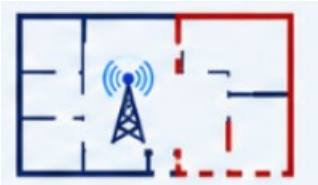
Future Directions



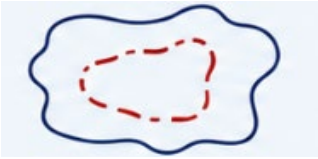
Outlook: this integrated architecture has not been jointly validated by the three papers.

The next step is not more physics terms, but adaptive coordination among multiple reliable priors.

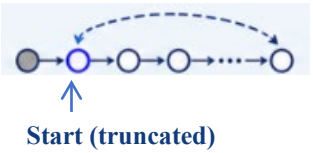
Key Takeaways and Q&A

1 

Geometry priors identify where propagation changes.

2 

Wave-equation priors identify what field structures must be preserved.

3 

Propagation priors determine how far diffusion needs to travel.

Code can be found at this link

WeChat group of radio map

Thanks You!

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